

CHAPTER 1 – INTRODUCTION

1.1 Introduction

Security Operations Centers (SOCs) are responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity incidents. With the increasing volume and complexity of security logs generated by modern systems, manual analysis has become time-consuming and resource-intensive. Security analysts often need to correlate information from multiple log sources, identify Indicators of Compromise (IOCs), map adversary behavior to standardized frameworks such as MITRE ATT&CK, assess incident severity, and prepare detailed incident reports. Performing these tasks manually can delay incident response and increase the likelihood of overlooking critical threats.

The AI SOC Copilot is designed to assist SOC analysts by automating significant portions of the incident analysis workflow. The application enables users to upload security logs, which are parsed and analyzed using Google Gemini Large Language Model (LLM). The system automatically extracts Indicators of Compromise (IOCs), classifies the detected threat, generates a risk score and severity level, maps the attack to the MITRE ATT&CK framework, and produces a comprehensive incident report with recommended containment and recovery actions.

The project combines artificial intelligence with cybersecurity best practices to reduce manual effort, improve consistency in threat analysis, and support faster incident response. Through an interactive web interface, analysts can visualize threat information, review attack timelines, inspect extracted IOCs, and export professional incident reports in PDF format.

1.2 Problem Statement

Modern organizations generate a large volume of security logs from endpoints, applications, authentication systems, and network devices. Manual analysis of these logs is repetitive, time-consuming, and highly dependent on the experience of security analysts. Identifying malicious activity, extracting relevant Indicators of Compromise (IOCs), correlating events, assigning appropriate severity levels, and preparing incident reports often require significant manual effort.

The objective of this project is to develop an AI-assisted Security Operations Center (SOC) Copilot capable of automating log analysis, threat classification, IOC extraction, MITRE ATT&CK mapping, severity assessment, and incident report generation, thereby improving the efficiency and effectiveness of security analysts.

1.3 Objectives

The primary objectives of the AI SOC Copilot are:

- Develop an intelligent web-based SOC assistant for cybersecurity incident analysis.
 - Accept uploaded security log files in supported formats.
 - Parse raw security logs into structured events.
 - Automatically extract Indicators of Compromise (IOCs) using Google Gemini.
 - Classify detected cybersecurity threats using AI.
 - Generate AI-based risk scores and severity levels.
 - Map detected attacks to the MITRE ATT&CK framework.
 - Generate detailed incident reports including executive and technical summaries.
 - Display analysis results through an interactive dashboard.
 - Export incident reports in PDF format.
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CHAPTER 2 – LITERATURE REVIEW

2.1 Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, investigating, and responding to cybersecurity incidents. SOC analysts continuously examine security events generated by operating systems, applications, networks, and security tools to identify suspicious activities. Modern SOC's increasingly rely on automation and artificial intelligence to improve response times and reduce analyst workload.

2.2 Indicators of Compromise (IOCs)

Indicators of Compromise (IOCs) are forensic artifacts that indicate the presence of malicious activity within a system or network. Common IOCs include IP addresses, malicious domains, URLs, email addresses, cryptographic file hashes, and vulnerability identifiers (CVEs). Extracting IOCs enables analysts to correlate incidents, identify attacker infrastructure, and support threat hunting activities.

2.3 MITRE ATT&CK Framework

The MITRE ATT&CK framework is a globally recognized knowledge base that documents adversary tactics, techniques, and procedures (TTPs). Mapping detected attacks to MITRE ATT&CK provides analysts with a standardized understanding of attacker behavior and supports consistent incident investigation and response.

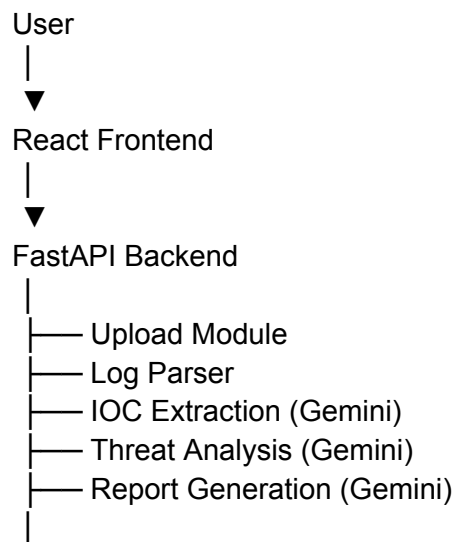
2.4 Large Language Models in Cybersecurity

Recent advances in Large Language Models (LLMs) have enabled AI systems to perform complex cybersecurity tasks such as log analysis, threat classification, IOC extraction, incident summarization, and security report generation. In this project, Google Gemini is utilized to analyze parsed security logs, extract structured threat intelligence, map attacks to MITRE ATT&CK techniques, calculate severity and risk scores, and generate professional incident reports.

CHAPTER 3 – SYSTEM ARCHITECTURE

3.1 Overall Architecture

The AI SOC Copilot follows a modular architecture consisting of frontend, backend, AI analysis services, and report generation components.



3.2 Technologies Used

Component	Technology
Frontend	React + Vite
Backend	FastAPI
AI Model	Google Gemini
Programming Language	Python
Styling	Tailwind CSS
Report Generation	PDF Export
Version Control	Git & GitHub

3.3 System Modules

The project consists of the following modules:

Upload Module

Allows users to upload security log files for analysis.

Log Parser

Processes uploaded logs and converts them into structured events.

IOC Extraction Module

Uses Google Gemini to identify IP addresses, domains, URLs, email addresses, hashes, and CVE identifiers.

Threat Analysis Module

Analyzes parsed logs and extracted IOCs to determine the attack type, severity, confidence score, risk score, and MITRE ATT&CK mapping.

Incident Report Module

Generates executive summaries, technical summaries, attack timelines, recommended containment actions, recovery steps, and analyst notes.

Dashboard

Displays all generated information including risk score, severity, MITRE mapping, IOC summary, incident timeline, and recommendations through an interactive interface.